Duty Cycle-Aware Battery Degradation Analytics for Electric Vehicle Fleet Applications

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1. Working Title

Duty Cycle-Aware Battery Degradation Analytics for Electric Vehicle Fleet Applications: *An Explainable Machine Learning Framework with Fleet Severity Indexing*

2. General Overview of the Research Area

Battery degradation in electric vehicles (EVs) has received considerable research attention in recent years, driven by the commercial imperative of reducing whole-life fleet costs and the policy requirement of the UK's Zero Emission Vehicle mandate (Department for Transport, 2023). A large body of studies has applied machine learning (ML) techniques to laboratory battery datasets to predict State of Health (SoH) and Remaining Useful Life (RUL). However, existing investigations have been confined largely to controlled cycling conditions that fail to reflect the real-world operational variability experienced by commercial EV fleets. The research has tended to focus on model performance accuracy rather than on the operational context that drives degradation — a distinction that matters greatly for fleet operators, OEMs, and battery management system (BMS) developers.

This research seeks to close that gap. The aim of this study is to develop a duty cycle-aware battery degradation prediction framework for EV fleet applications, integrating drive cycle analysis, multi-model ML benchmarking, and SHAP-based explainability within a unified, data-driven pipeline. A key novel contribution is the Fleet Severity Index (FSI) — a quantitative scoring framework that ranks duty cycle impact on battery health across a simulated fleet population. The study is quantitative and deductive in design, fully simulation-based using open-source datasets, with no physical experimentation or human participants involved. Implementation will be carried out in Python and MATLAB.

The researcher brings over four years of professional experience in EV battery testing, BMS validation, and powertrain engineering, providing direct industry grounding to the research questions pursued here.

3. Identification of the Relevant Literature

Although considerable research has been devoted to ML-based SoH estimation, rather less attention has been paid to the influence of real-world drive cycle behaviour on battery ageing trajectories. The literature can be grouped around three themes relevant to this study.

ML-based degradation and RUL prediction. Ensemble methods such as Random Forest and XGBoost have shown strong performance for SoH prediction on tabular battery data (MDPI, 2024). Zhang et al. (2024) propose a scientific ML framework combining domain knowledge with neural networks for short and long-term degradation forecasting, while Yu et al. (2024) demonstrate that FEEMD-LSTM-TAM models achieve sub-one-cycle RUL error on NASA and CALCE datasets. LSTM networks remain the standard for sequential degradation modelling, as established by the foundational contribution of IEEE (2018). However, few studies benchmark multiple model architectures simultaneously against drive-cycle-informed feature sets.

Explainability in battery AI. Si et al. (2025) apply SHAP analysis to EIS-derived features for capacity estimation in lithium metal batteries, demonstrating the diagnostic utility of explainability beyond prediction accuracy. Wang et al. (2025) combine GBM and LSTM with both SHAP and LIME, finding SHAP provides more stable attributions for degradation tasks. Despite this progress, SHAP has not yet been systematically applied to drive-cycle-derived feature sets in a fleet context.

Drive cycle impact and fleet-level degradation. Jiang et al. (2025) map battery capacity fade across diverse driving and environmental conditions, establishing operational variables as meaningful predictors of degradation. The Geotab EV Battery

Health study (2026), drawing on data from 22,700 vehicles, identifies high daily utilisation and DC fast charging as dominant fleet stressors, with average degradation of 2.3% per year. No study identified in this review proposes a structured Fleet Severity Index to quantify and rank duty cycle exposure across a fleet population — the central gap this research will address.

4. Key Research Questions

Applying Kipling's research framework (What, Why, When, How, Where, Who) to scope the study:

1. **WHAT:** To what extent can duty-cycle-derived operational stress indicators improve the accuracy and interpretability of ML-based battery degradation prediction in EV fleet applications?
2. **WHY:** Because existing frameworks ignore operational context, leaving fleet operators without actionable battery intelligence for maintenance decisions.
3. **HOW:** Through a quantitative ML pipeline using open-source datasets, drive cycle analysis, SHAP explainability, and FSI construction — implemented in Python and MATLAB.
4. **WHERE:** The research is located in the EV engineering and battery health management domain, with applied relevance to UK fleet operators and UK ZEV mandate policy.
5. **WHEN:** June to September 2026, within the EG7030 dissertation module at UEL.

Subsidiary questions include:

- Which ML model — RF, XGBoost, Gradient Boosting, SVM, or LSTM — delivers the most accurate degradation prediction when trained on drive-cycle-informed features?
- Which operational variables, identified via SHAP analysis, contribute most significantly to capacity fade across WLTP, UDDS, FTP-75, HWFET, and NEDC profiles?
- Can a Fleet Severity Index serve as a practical tool for ranking and prioritising battery maintenance interventions in an EV fleet?

5. Methodology

This study adopts a quantitative, deductive research design. Data will be collected from three open-source battery degradation datasets — the NASA Prognostics Battery Dataset, the CALCE collection (University of Maryland), and the Oxford Battery

Degradation Dataset — supplemented by fleet operational context from the Geotab and NREL datasets. These sources provide the lithium-ion cycling data needed to train and evaluate the ML models.

The methodology proceeds through six stages: (1) data acquisition and pre-processing — cleaning, normalisation, interpolation, and cycle segmentation; (2) drive cycle and operational stress analysis — extracting current demand profiles, C-rate variability, acceleration/deceleration behaviour, and regenerative braking intensity from WLTP, UDDS, FTP-75, HWFET, NEDC, and urban stop-go profiles; (3) feature engineering — deriving SoH, internal resistance growth, coulombic efficiency, equivalent full cycles, RMS current stress, and thermal exposure indicators; (4) ML model benchmarking — training and comparing Random Forest, XGBoost, Gradient Boosting, SVM, and LSTM for fault classification, degradation prediction, and RUL estimation; (5) SHAP explainability analysis — identifying dominant degradation drivers across models and drive cycle profiles; and (6) FSI construction and fleet maintenance framework design — producing a ranked battery severity score and predictive maintenance schedule.

Model performance will be evaluated using Accuracy, Precision, Recall, F1-score, RMSE, MAE, and k-fold cross-validation. Python (scikit-learn, XGBoost, TensorFlow/Keras, SHAP) and MATLAB will be the primary tools.

6. Timescale and Research Planning

Phase	Period	Key Activities
1 — Setup & Literature	May–Jun 2026	Supervisor meetings, literature review, dataset download, Python/MATLAB setup
2 — Data Preparation	Early Jun 2026	EDA, cleaning, normalisation, drive cycle integration, feature engineering
3 — Model Development	Mid Jun–Jul 2026	ML model training, hyperparameter tuning, cross-validation (RF, XGBoost, LSTM)
4 — Framework Design	Late Jul 2026	FSI construction, SHAP analysis, maintenance decision framework
5 — Results & Analysis	Early Aug 2026	Results, visualisations, discussion, limitations, UK policy contextualisation
6 — Writing & Submission	Aug–Sep 2026	Full dissertation (~12,500 words), supervisor review, Turnitin submission

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